

Chapter 1: Introduction and Basic Concepts

- $E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$. Energy conservation principle (First Law): the net change in a system's total energy equals the energy input minus energy output.
- $E_{\text{total}} = U + KE + PE$, with $KE = \frac{1}{2}mV^2$ and $PE = mgz$. Total energy is composed of internal energy U (molecular energy), kinetic energy, and potential energy.
- $P_{\text{abs}} = P_{\text{atm}} + P_{\text{gage}}$. Relationship between absolute, atmospheric, and gage pressure. Gage pressure is measured relative to the local atmosphere, so absolute pressure is gage pressure plus atmospheric pressure.
- $P = P_0 + \rho gh$. Pressure in a static fluid increases with depth h (due to weight of fluid). This formula (hydrostatic equation) gives pressure at depth, where P_0 is pressure at a reference level.
- $T_{(K)} = T_{(^{\circ}C)} + 273.15$; $T_{(^{\circ}F)} = 1.8 T_{(^{\circ}C)} + 32$. Temperature conversion formulas between Celsius, Kelvin, and Fahrenheit scales. $273.15 \text{ K} = 0 \text{ }^{\circ}\text{C}$, and $32 \text{ }^{\circ}\text{F} = 0 \text{ }^{\circ}\text{C}$ with a 1.8 scale factor.
- $\rho = \frac{m}{V}$, $v = \frac{1}{\rho}$. Definition of density ρ (mass per unit volume) and specific volume v (volume per unit mass).
- $\gamma = \rho g$. Specific weight (useful for hydrostatics).
- Manometers / hydrostatics (gage pressure): $p_{\text{gage}} = \rho gh$. For a differential manometer between two fluids, $\Delta p = (\rho_2 - \rho_1)gh$. Example with mercury vs water: $\Delta p = (13.6 - 1)\rho_{\text{water}}gh = 12.6\rho_{\text{water}}gh$.
- Same-level rule: at the same elevation in a connected, static fluid, pressures are equal. Use it to relate columns in U-tubes/multifluids.

Chapter 2: Energy, Energy Transfer & General Energy Analysis

- $\Delta E = Q - W$. First Law for a closed system: the change in total energy (internal + kinetic + potential) equals heat Q added to the system minus work W done by the system. (Sign convention: Q in is positive, W out is positive).
- $\Delta U = Q - W_{\text{other}} - W_b$. Energy balance for a closed system including different work forms. For example, $W_b = \int P dV$ is boundary (PdV) work, and W_{other} could be electric or shaft work.
- $W = \int F ds = \int P dV$. Definition of work: e.g. boundary work is the area under the P - V curve of a process. Positive if volume expands (work done by system).
- $\eta_{\text{energy conv}} = \frac{\text{desired output}}{\text{energy input}}$. General definition of energy conversion efficiency (e.g. for an engine or any energy device). It is the ratio of useful energy output to the energy input.
- $w_{\text{flow}} = Pv$. Flow work per unit mass to push fluid into a control volume.
- $e_{\text{mech}} = Pv + \frac{V^2}{2} + gz$. Specific mechanical energy of a flowing fluid (flow, kinetic, potential).
- $e_{\text{flow}} = h + \frac{V^2}{2} + gz$, $\dot{E}_{\text{mass}} = \dot{m} \left(h + \frac{V^2}{2} + gz \right)$. Energy carried with mass flow.

Chapter 3: Properties of Pure Substances

- $x = \frac{m_{\text{vapor}}}{m_{\text{total}}}$. Quality of a saturated liquid-vapor mixture: the mass fraction that is vapor. $x = 0$ for saturated liquid, $x = 1$ for saturated vapor.
- $y = y_f + x(y_g - y_f)$. Property in a saturated mixture: specific volume v , internal energy u , enthalpy h , entropy s , etc., of a mixture are the weighted average of saturated liquid (f) and saturated vapor (g) values: $y = y_f + x(y_g - y_f)$.
- $h = u + Pv$. Definition of enthalpy (per unit mass). For a simple compressible substance, enthalpy combines internal energy and flow work (Pv). Enthalpy is useful for processes at constant pressure and in open systems.
- $Pv = RT$. Ideal gas equation of state (per unit mass form). Relates pressure P , specific volume v , and temperature T for ideal gases. R is the specific gas constant ($R = R_u/M$ where M is molar mass).
- $Z = \frac{Pv}{RT}$. Compressibility factor, a correction factor for real gases. $Z = 1$ for ideal gases; deviation of Z from unity indicates how much a real gas deviates from ideal gas behavior.

Chapter 4: Energy Analysis of Closed Systems

- $W_b = \int_{V_1}^{V_2} P dV$. Boundary work (expansion or compression work) for a quasi-equilibrium process, equal to the area under the P - V curve. For common processes: isobaric $W_b = P(V_2 - V_1)$; isothermal (ideal gas) $W_b = P_1 V_1 \ln(V_2/V_1)$; polytropic $W_b = \frac{P_2 V_2 - P_1 V_1}{1-n}$ (if $n \neq 1$).
- $Q - W = \Delta U + \Delta KE + \Delta PE$. Comprehensive energy balance for a closed system. The net heat input minus work output equals the change in the system's total energy (including internal, kinetic, and potential energy).
- $c_v = \left(\frac{\partial u}{\partial T} \right)_V$, $c_p = \left(\frac{\partial h}{\partial T} \right)_P$. Definitions of specific heats at constant volume and constant pressure. c_v is the change in internal energy per unit mass per degree at constant volume, and c_p is the change in enthalpy per unit mass per degree at constant pressure.
- $\Delta u \approx c_v(T_2 - T_1)$, $\Delta h \approx c_p(T_2 - T_1)$. For ideal gases (assuming constant specific heats), internal energy change depends only on temperature change (and similarly for enthalpy change). This is a useful approximation for many engineering calculations.
- $c_p - c_v = R$. Ideal gas specific heat relation (Mayer's relation). The difference between specific heat at constant pressure and constant volume equals the gas constant R for ideal gases.
- Spring-loaded piston (linear P - V): $W_b = \frac{P_1 + P_2}{2} (V_2 - V_1)$. Area of the trapezoid when P varies linearly with V .

Chapter 5: Mass and Energy Analysis of Control Volumes

- $\sum m_{\text{in}} - \sum m_{\text{out}} = \frac{dm_{\text{sys}}}{dt}$. Mass conservation for a control volume. In steady state, mass flow in equals mass flow out. For transient (unsteady) processes, any imbalance in mass flow causes a change in the mass of the control volume.
- $\frac{dE_{\text{cv}}}{dt} = \sum \dot{Q}_{\text{in}} - \sum \dot{Q}_{\text{out}} + \sum \dot{m} \left(h + \frac{V^2}{2} + gz \right)_{\text{in}} - \sum \dot{m} \left(h + \frac{V^2}{2} + gz \right)_{\text{out}}$. General energy rate balance for a control volume (open system). This accounts for heat, work, and energy transport via mass flow. For steady-state operation, $dE_{\text{cv}}/dt = 0$ and the equation simplifies accordingly.

- Steady-flow device applications (adiabatic idealizations):
 - Nozzle/Diffuser: $h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$. (No work, adiabatic flow device, converts enthalpy and kinetic energy.)
 - Turbine: $W_{\text{out}} = h_1 - h_2$. (Extracts work by dropping enthalpy of the flow.)
 - Pump/Compressor: $W_{\text{in}} = h_2 - h_1$. (Work input raises fluid enthalpy/pressure.)
 - Throttle/Valve: $h_2 \approx h_1$. (Pressure drop without work; enthalpy stays nearly constant during a throttling process.)
 - Mixing chamber (adiabatic, no shaft work):
$$\sum \dot{m}_{\text{in}} h_{\text{in}} = \sum \dot{m}_{\text{out}} h_{\text{out}}, \quad \sum \dot{m}_{\text{in}} = \sum \dot{m}_{\text{out}}, \quad \Delta e_K \approx \Delta e_P \approx 0.$$
 - Heat exchanger (treat streams separately): $\dot{Q}_{\text{cold}} = \dot{m} \Delta h_{\text{cold}}$, $\dot{Q}_{\text{hot}} = \dot{m} \Delta h_{\text{hot}}$, and $-\dot{Q}_{\text{hot}} = \dot{Q}_{\text{cold}}$; $\Delta e_K, \Delta e_P \approx 0$, no shaft work.
 - Throttle / valve (isenthalpic): $h_2 \approx h_1$ with $\dot{Q} \approx \dot{W} \approx 0$, $\Delta e_K, \Delta e_P$ small relative to Δh .

Symbols, Units & Notation

Thermodynamic Properties

Symbol	Name	SI unit	Notes
T	Temperature	K	$T[\text{K}] = T[^\circ\text{C}] + 273.15$
P	Pressure (absolute)	Pa (=N/m ²)	$P_{\text{abs}} = P_{\text{atm}} + P_{\text{gage}}$
P_{atm}	Atmospheric pressure	Pa	Local ambient pressure
P_{gage}	Gauge pressure	Pa	Often from manometer: $P_{\text{gage}} = \rho gh$
ρ	Density	kg/m ³	$\rho = 1/v$
v	Specific volume	m ³ /kg	$v = 1/\rho$
u	Specific internal energy	kJ/kg	Ideal gas: $u = u(T)$
U	Internal energy (total)	kJ	$U = mu$
h	Specific enthalpy	kJ/kg	$h = u + Pv$ (ideal gas: $h = h(T)$) $H = mh$
H	Enthalpy (total)	kJ	
s	Specific entropy	kJ/(kg · K)	From tables/relations
S	Entropy (total)	kJ/K	$S = ms$
x	Quality (vapor mass frac.)	–	$x = m_{\text{vapor}}/m$; $y = y_f + x(y_g - y_f)$
R	Gas constant (specific)	kJ/(kg · K)	$R = R_u/M$; air ≈ 0.287
R_u	Universal gas constant	J/(mol · K)	8.314
Z	Compressibility factor	–	$Z = Pv/(RT)$
c_p	Spec. heat at const P	kJ/(kg · K)	$c_p - c_v = R$
c_v	Spec. heat at const V	kJ/(kg · K)	
γ	Specific weight	N/m ³	$\gamma = \rho g$ (hydrostatics)

Mechanics & Flow

Symbol	Name	SI unit	Notes
m	Mass	kg	
$\dot{m}$	Mass flow rate	kg/s	$\dot{m} = \rho \bar{v} A$
V	Speed (magnitude)	m/s	Flow/stream speed; vector often $\vec{V}$
$\bar{v}$	Mean (bulk) velocity	m/s	Cross-section average
$\dot{V}$	Volumetric flow rate	m ³ /s	$\dot{V} = \bar{v} A$
A	Cross-sectional area	m ²	
z	Elevation	m	Datum for potential energy
g	Gravitational accel.	m/s ²	$g \approx 9.81$
e_{mech}	Spec. mechanical energy	J/kg	$e_{\text{mech}} = \frac{P}{\rho} + \frac{V^2}{2} + gz$
e_{flow}	Spec. flow energy	J/kg	$e_{\text{flow}} = h + \frac{V^2}{2} + gz$
W	Work (total)	kJ	+/- by your sign convention
w	Specific work	kJ/kg	$w = W/m$
$\dot{W}$	Power (rate of work)	kW	$\dot{W} = dW/dt$
Q	Heat transfer (total)	kJ	+/- by your sign convention
q	Specific heat transfer	kJ/kg	$q = Q/m$
$\dot{Q}$	Heat transfer rate	kW	$\dot{Q} = dQ/dt$
W_b	Boundary work (total)	kJ	$W_b = \int P dV$
w_b	Boundary work (specific)	kJ/kg	$w_b = \int P dv$
τ	Torque	N · m	Shaft: $\dot{W}_{sh} = \tau \omega$
ω	Angular speed	rad/s	$\omega = 2\pi n$
n	Rotational speed	s ⁻¹ (Hz)	Often given in rpm
k	Spring constant	N/m	$W_{\text{spring}} = \frac{1}{2} k(x_2^2 - x_1^2)$

Devices, Cycles & Performance

Symbol	Name	SI unit	Notes
η_{th}	Thermal efficiency	–	$\eta_{\text{th}} = W_{\text{net,out}}/Q_H = 1 - Q_L/Q_H$
COP_R	COP (refrigerator)	–	$\text{COP}_R = Q_L/W_{\text{in}}$
COP_{HP}	COP (heat pump)	–	$\text{COP}_{HP} = Q_H/W_{\text{in}} = 1 + \text{COP}_R$
HV	Heating value (fuel)	kJ/kg (or kJ/kmol)	Use LHV/HHV as specified
n	Polytropic exponent	–	$PV^n = \text{const}$ along path

Subscripts, Superscripts & Notation

Marking	Meaning	Example	Notes
($\dot{}$)	Time rate	$\dot{m}$, $\dot{Q}$	“per unit time”
Lowercase vs uppercase	Specific vs total	u vs $U = mu$	Per-unit-mass vs total
Overbar	Cross-sec. average	$\bar{v}$	Mean (bulk) velocity
$\Delta()$	Change between states	$\Delta U = U_2 - U_1$	Finite difference
1, 2	State indices	T_1 , T_2	Inlet/exit or initial/final
in, out	Stream tags	h_{in}	Control-volume book-keeping
f	Sat. liquid	v_f , h_f	At $x = 0$
g	Sat. vapor	v_g , h_g	At $x = 1$
fg	Difference $g - f$	$h_{fg} = h_g - h_f$	Latent jump
sat	Saturated	P_{sat} , T_{sat}	Sat. line properties
abs, gage, atm	Pressure types	P_{abs}	Absolute, gauge, atmospheric

Handy Constants

Quantity	Symbol	Value	Notes
Gravitational acceleration	g	9.81 m/s ²	Standard gravity
Univ. gas constant	R_u	8.314 J/(mol · K)	= 8.314 kJ/(kmol · K)
Air gas constant	R_{air}	0.287 kJ/(kg · K)	Ideal-gas calcs
Water specific weight (approx)	γ	≈ 9.81 kN/m ³	At 4°C